

N1 Data Ops Take Home Challenge

Description

N1 Health just brought on a new client. You have been assigned to this client's team and are tasked with investigating the customer's data and providing some high-level analysis.

Challenge

Several member rosters have been loaded into a SQLite database. Your task is to take these member rosters and combine them to get a full view of the client's membership. Then, using the complete membership, create the necessary summary statistics to be used for further analysis.

1. Using Python and SQLite, combine the member rosters and create the requested tables and summary statistics (see below). Put this script on GitHub or other repository for review.
2. Draft an email to the "project manager" explaining your process, and the answers to their questions.
3. Create a short presentation to demonstrate your work.

If any requirements are not clear, feel free to make any assumptions needed to complete the challenge. Be sure to document those assumptions in your email. Specific questions to the team are also welcome.

Data

You have been provided with a SQLite database that contains 5 member roster tables from the client, as well as one table with model scores broken up by zip code.

Final Deliverables

- A. **Create a Python script** that fulfills the following requirements:
- a. **Import** the `sqlite3` package to read and query the data in the provided SQLite database
 - b. **Create a table** called `std_member_info` that:
 - Combines and standardizes the data for all members who had active eligibility **this year**.
 - Has a cardinality of 1 row per member.
 - Includes the following columns:

- member_id	- main_address	- payer
- member_first_name	- city	- eligibility_start_date
- member_last_name	- state	- eligibility_end_date
- date_of_birth	- zip_code	
 - c. **Calculate and display** the answers to the following questions:
 1. *How many distinct members were eligible in April 2025?*
 2. *How many members were included more than once?*
 3. *What is the breakdown of members by payer?*
 4. *How many members live in a zip code with a "Food access score" less than 2?*
 5. *What is the average "Social isolation score" for all members?*
 6. *Which members live in the zip code with the highest "Algorex SDOH composite score"?*
- B. **Draft an email** to the project manager explaining the steps you took, any issues you discovered with the data, and the requested outputs. Remember that the project manager is non-technical.
- C. **Create a presentation** (no more than 5 slides) explaining your approach and the results of your work. Expect to present for about 15 minutes, including an introduction, a summary of the results, and a short walkthrough of your code. Additionally, consider the following scenario:

"Starting next month, we will begin to receive a new member roster from the client every 2 weeks. What changes would you need to make to your script to accommodate this new data?"

Please submit your scripts and any supporting materials at least 1 day prior to your presentation date.